ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **How do Organisms Reproduce?** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** at a definite plane related to the whip-like flagellum	Leishmania has a definite orientation of division; Amoeba divides in any plane.
2	**(c)** pollen grain	Pollen grains carry the male gametes.
3	**(d)** testes	Testes are the male sex organs.
4	**(d)** Plasmodium	Plasmodium divides into many daughter cells at once.
5	**(c)** ovary	After fertilisation the ovary ripens into fruit.
6	**(d)** sperm formation needs a temperature lower than body temperature	A lower temperature is essential for sperm production.
7	**(b)** budding	A bud grows out as an outgrowth and detaches.
8	**(c)** seed	The fertilised ovule becomes the seed.
9	**(c)** ovaries	Ovaries produce eggs.
10	**(c)** spores	Sporangia release many spores.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Variations help some individuals survive when the environment changes (1); This ensures the survival of the species over generations (1).	1+1
12	New plants are obtained from vegetative parts such as root, stem or leaf (1); Rose, Bryophyllum or sugarcane (1).	1+1
13	Provides nutrition and oxygen to the embryo (1); Removes waste from the embryo to the mother's blood (1).	1+1
14	Binary fission: a cell divides into two daughter cells (1); Multiple fission: a cell divides into many daughter cells at once (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	A bud develops as an outgrowth due to repeated cell division (1); The bud grows into a tiny individual with its own mouth and tentacles (1); It detaches when mature and becomes an independent Hydra (1).	1+1+1
16	Anther produces pollen grains (1); Stigma receives pollen (1); Ovary contains ovules and becomes fruit (1).	1+1+1
17	Boys: growth of facial hair, voice change (1); Girls: breast development, onset of menstruation (1); Both: hair under the arm and genital area, sexual maturity (1).	1+1+1
	OR: DNA carries the information for building body design (1); Copies of DNA pass the same body design to the offspring (1); Slight inaccuracies in copying create variation (1).	1+1+1
18	Zygote divides to form the embryo in the ovule (1); The ovule develops a tough coat and becomes a seed (1); The ovary grows and ripens to form the fruit; petals, sepals and stamens wither (1).	1+1+1
	OR: The embryo is embedded in the uterine wall (1); The placenta is a disc-like tissue with villi (1); Nutrients and oxygen pass from mother's blood to the embryo, and wastes go back (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Fission (binary and multiple): Amoeba, Plasmodium (1); Fragmentation: Spirogyra (1); Regeneration: Planaria (1); Budding: Hydra, yeast (1); Spore formation: Rhizopus (1).	1+1+1+1+1
	OR: A pair of ovaries produce eggs (1); The egg enters the oviduct (fallopian tube) (1); Fertilisation occurs in the oviduct (1); Zygote moves and is implanted in the uterus (1); Uterus opens into the vagina through the cervix (1).	1+1+1+1+1
20	Nucleus elongates and divides (1); The cytoplasm then constricts and splits (1); Two daughter Amoebae are formed (1); Division can occur in any plane (1); Leishmania divides in a definite orientation with respect to the whip-like structure (1).	1+1+1+1+1

	OR: Zygote divides to form the embryo (1); Embryo is implanted in the thick uterine lining (1); Placenta develops and nourishes the embryo (1); Embryo grows to a foetus in about nine months (1); Birth occurs by contractions of the uterine muscles (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that Plasmodium reproduces by binary fission; it undergoes **multiple** fission, whereas Amoeba shows binary fission.
- Using 'fertilisation' for pollination; pollination is transfer of pollen, fertilisation is fusion of male and female gametes.
- Saying that fertilisation takes place in the uterus; it occurs in the oviduct and the zygote is implanted in the uterus.
- Confusing regeneration with reproduction by fragmentation; regeneration uses specialised cells to rebuild a missing part, e.g. Planaria, Hydra.
- Stating that asexual reproduction produces absolutely identical offspring; tiny variations are produced by errors in DNA copying.
- Believing that the testes lie inside the abdomen; they are in the scrotum because sperm formation requires a lower temperature.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **How do Organisms Reproduce?** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** pollination	This is the definition of pollination.
2	**(c)** oviduct (fallopian tube)	The egg meets the sperm in the oviduct.
3	**(d)** regeneration	Specialised cells form new tissues for a missing part.
4	**(b)** cross-pollination	Pollen comes from another plant.
5	**(b)** uterus	The embryo is embedded in the thick uterine lining.
6	**(c)** fragmentation	The filament breaks into pieces that grow into new filaments.
7	**(c)** style	It carries the male gametes to the ovule.
8	**(d)** placenta	The placenta lets nutrients and oxygen pass to the embryo.
9	**(d)** leaves	Buds in the notches of leaf margins develop into plantlets.
10	**(d)** pistil (carpel)	Pistil contains stigma, style and ovary.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Plants bear flowers and fruits earlier (1); Seedless plants like banana can be grown (1).	1+1
12	The thick lining is not needed (1); It breaks and comes out through the vagina with blood as menstruation (1).	1+1
13	Growth of a complete organism from a body part (1); Planaria or Hydra (1).	1+1
14	Male: stamen (anther and filament) (1); Female: pistil (stigma, style, ovary) (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Hyphae bear blob-like sporangia (1); Sporangia contain spores that are tiny and covered by thick walls (1); Spores germinate into new Rhizopus on getting moisture (1).	1+1+1
16	Self: pollen reaches the stigma of the same flower or plant (1); Cross: pollen reaches a flower of a different plant (1); Cross-pollination gives more variation (1).	1+1+1
17	Bleeding from the vagina about every month (1); The uterus prepares a thick lining for the embryo (1); If no egg is fertilised, the lining breaks down and is shed (1).	1+1+1
	OR: Fragmentation: the body breaks into pieces that grow into new individuals (1); Regeneration: specialised cells grow into missing parts (1); Example: Spirogyra for fragmentation; Planaria for regeneration (1).	1+1+1
18	A seed has an embryo covered by a seed coat with cotyledons (1); Plumule develops into the shoot (1); Radicle develops into the root (1).	1+1+1
	OR: The tail helps them swim towards the egg (1); They have to travel a long distance through the female tract (1); The head contains the genetic material (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	New plants come from roots, stems or leaves (1); Bryophyllum: leaf buds form plantlets (1); Rose or sugarcane: cuttings of stem (1); Plants flower and fruit earlier (1); Seedless plants like banana can be grown and the plants are genetically identical (1).	1+1+1+1+1
	OR: Cycle of changes in the uterus of about 28 days (1); Ovary releases an egg each month (1); Uterine lining thickens in preparation (1); If no fertilisation occurs the lining breaks (1); Blood and tissue are shed as menstruation lasting 2 to 8 days (1).	1+1+1+1+1
20	Sepals form the outer protective whorl (1); Petals are coloured to attract insects (1); Stamens: anther and filament, produce pollen (1); Pistil: stigma, style, ovary (1); Ovary contains ovules; unisexual and bisexual flowers differ (1).	1+1+1+1+1
	OR: DNA copying is never perfectly accurate (1); Sexual reproduction combines DNA of two parents (1); Gametes carry different combinations of genes (1); Offspring show new combinations (1); Variations help species survive changes in the environment (1).	1+1+1+1+1

Mistakes to avoid:

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- Saying that fertilisation takes place in the uterus; it occurs in the oviduct and the zygote is implanted in the uterus.
- Confusing regeneration with reproduction by fragmentation; regeneration uses specialised cells to rebuild a missing part, e.g. Planaria, Hydra.
- Stating that asexual reproduction produces absolutely identical offspring; tiny variations are produced by errors in DNA copying.
- Believing that the testes lie inside the abdomen; they are in the scrotum because sperm formation requires a lower temperature.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **How do Organisms Reproduce?** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** breaks down and leaves as menstruation	The lining is shed with blood.
2	**(c)** Rose	Rose is propagated by cuttings (stem).
3	**(c)** fertilisation	Fusion forms the zygote.
4	**(d)** 28 days	The usual cycle is about a month.
5	**(d)** genetically similar to the parent	There is only one parent and DNA is copied.
6	**(c)** plumule	Plumule forms the shoot, radicle forms the root.
7	**(b)** provides fluid and nutrition to the sperms	It forms seminal fluid.
8	**(d)** help it survive environmental changes	Variants may survive when conditions change.
9	**(b)** water, air and suitable temperature	Seeds need these three to sprout.
10	**(a)** breast development and start of menstruation	Girls develop breasts and begin menstruation.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sperm formation needs a temperature lower than body temperature (1); The scrotum keeps the testes cooler (1).	1+1
12	Rhizopus produces a large number of spores (1); The spores germinate quickly in warm and moist conditions (1).	1+1
13	Pollination: transfer of pollen from anther to stigma (1); Fertilisation: fusion of male and female gametes (1).	1+1
14	It grows through the style to the ovary (1); It carries male gametes to the female gamete in the ovule (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Hydra regenerates from cut pieces (1); Regeneration is carried out by specialised cells; most organisms cannot reproduce this way (1); Reproduction is a general process of making new individuals (1).	1+1+1
16	Testis produces sperms and testosterone (1); Vas deferens carries sperms (1); Urethra is the common passage for sperms and urine (1).	1+1+1
17	Sexual needs two parents/gametes; asexual needs one parent (1); Sexual offspring show variation; asexual offspring are nearly identical (1); Sexual reproduction is slower; asexual is rapid (1).	1+1+1
	OR: Their body is made of specialised cells arranged into organs (1); Such organisation cannot be broken and rebuilt easily (1); So they use specialised reproductive organs (1).	1+1+1
18	Ovary produces eggs and female hormones (1); Oviduct carries the egg; fertilisation takes place here (1); Uterus is where the embryo develops (1).	1+1+1
	OR: Papaya and watermelon have unisexual flowers (1); Hibiscus has bisexual flowers with both stamens and pistil (1); Unisexual flowers carry either stamens or pistil only (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Pollen transferred from anther to stigma (1); Pollen tube grows through the style (1); Male gamete fuses with the female gamete in the ovule (fertilisation) (1); Zygote forms the embryo; ovule becomes the seed (1); Ovary becomes the fruit (1).	1+1+1+1+1
	OR: Number of parents: one vs two (1); Gametes: not involved vs involved (1); Variation: very little vs more (1); Speed: fast vs slower (1); Examples: Hydra budding vs flowering plants and humans (1).	1+1+1+1+1
20	Testes in the scrotum produce sperms and testosterone (1); Sperms pass into the vas deferens (1); Prostate and seminal vesicles add secretions (1); Sperms leave through the urethra and penis (1); Testes lie outside the body for a lower temperature (1).	1+1+1+1+1
	OR: Seed coat, cotyledons and embryo (plumule and radicle) (1); Seed takes up water and swells (1); Radicle emerges first and forms the root (1); Plumule forms the shoot (1); Cotyledons supply food till the plant makes its own (1).	1+1+1+1+1

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